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ICE 6.1



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5/8

1. (1 mark) AES is yet another encryption system. Is it symmetric or asymmetric cryptosystem?

- **symmetric(correct)**
- asymmetric
- public-key crypto system
- None of the above

2. (1 mark) AES is expected to be far more secure than DES. Is it true or false? And why?

- True, since it is now used in military and civil applications by all countries
- **True, since no single country owns/controls its design (correct)**
- False
- None of the above

3. (1 mark) AES comes in different flavours, viz. no. of rounds and key size. In particular the number of rounds is:

- **10, 12, 14 (correct)**
- 10, 12
- 10

4. (2 marks) Consider the design of the S-box and its inverse S^{-1} , or look-up tables corresponding to them. One property that the S box satisfies is that there are no self-inverses, viz. $S(a) \neq S^{-1}(a)$. From the look-up tables for $S(\cdot)$ and $S^{-1}(\cdot)$ which one of those given below shows that this is indeed the case.

- $S(64)=43$; $S^{-1}(46)=98$
- **$S(64)=43$; $S^{-1}(64)=8C$ (correct)**
- $S(64)=43$; $S^{-1}(43)=64$
- None of the above

5. (3 marks) Let $M1$, $M2$ and $M3$ be the state of the text within a given round but after the operations (i) "substitute byte", (ii) "shift rows", (iii) "mix columns". To be very specific, which are all the elements in $M1$ that contribute to determination of $M3(2,2)$.



- **M1(1,2), M1(2,3), M1(3,4), M1(4,1) (correct)**
- M1(1,2), M1(2,2), M1(3,2), M1(4,2)
- M1(1,1), M1(2,2), M1(3,3), M1(4,4)

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